Vitamin	Sources	Functions	Deficiency Disease
Vitamin B_{12} (Cobalamin)	Egg, Meat, Fish	Co-enzyme in amino acid metabolism, Red blood cells maturation	Pernicious Anaemia
Vitamin C (Ascorbic acid)	Citrus fruits (Orange, Lemon etc…),Tomato, Amla, Leafy Vegetables	Coenzyme in Antioxidant, building of collagen	Scurvy (bleeding gums)
Vitamin D Cholecalciferol(D3), Ergocalciferol (D2)	Fish liver oil, Milk, Egg yolk, (exposure to sunlight)	Absorption and maintenance of calcium	Rickets (children), Osteomalacia (adults)
Vitamin E (Tocopherols)	Cotton seed oil, Sun flower oil, wheat germ oil, Vegetable oils	Antioxidant	muscular dystrophy (muscular weakness) and neurological dysfunction
Vitamin K (Phylloquinone& Menaquinones)	Green leafy vegetable, soybean oil, tomato	Blood clotting	Increased blood clotting time, Haemorrhagic diseases

14.5 Nucleic acids

The inherent characteristics of each and every species are transmitted from one generation to the next. It has been observed that the particles in nucleus of the cell are responsible for the transmission of these characteristics. They are called chromosomes and are made up of proteins and another type of biomolecules called nucleic acids. There are mainly two types nucleic acids, the deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). They are the molecular repositories that carry genetic information in every organism.

14.5.1 Composition and structure of nucleic acids

Nucleic acids are biopolymers of nucleotides. Controlled hydrolysis of DNA and RNA yields three components namely a nitrogenous base, a pentose sugar and phosphate group.

Nitrogen base

These are nitrogen containing organic compounds which are derivatives of two parent

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